

# Smart Factory Daily KPI Report

Report Date: 2026-03-31

## 1. Executive Summary

This automated daily report summarizes production performance, OEE, yield, UPH, downtime, and alert status for 2026-03-31. The report is generated from simulated MES-style production data.

## 2. Daily KPI Summary

| Metric                | Value  |
|-----------------------|--------|
| Total Planned Output  | 11,436 |
| Total Actual Output   | 10,111 |
| Total Good Units      | 9,724  |
| Total Defective Units | 387    |
| Total Downtime Hours  | 11.83  |
| Availability          | 91.78% |
| Performance           | 88.41% |
| Quality               | 96.17% |
| OEE                   | 78.04% |
| Yield                 | 96.17% |
| Defect Rate           | 3.83%  |
| UPH                   | 76.50  |
| Downtime Rate         | 8.22%  |

## 3. Operational Diagnostics

| Item                      | Result                  |
|---------------------------|-------------------------|
| Top Downtime Reason       | Preventive Maintenance  |
| Top Downtime Minutes      | 169 minutes             |
| Worst Machine by Downtime | M06                     |
| Worst Machine Downtime    | 152 minutes             |
| Alert Status              | Below Average UPH Alert |

## 4. Alert Interpretation

The system triggered the following alert(s): Below Average UPH Alert. Further investigation should focus on OEE loss, downtime drivers, yield performance, and machine-level constraints.

## 5. Notes

This report is generated automatically by the Streamlit-based Smart Factory KPI Monitoring and Reporting Automation System. The dataset is simulated for portfolio and demonstration purposes.